

Ex. No. 5: MIMO System Simulation with BER Analysis

AIM

To simulate a **MIMO (Multiple Input Multiple Output)** wireless communication system and analyze its performance in terms of **Bit Error Rate (BER)**, **Signal-to-Noise Ratio (SNR)**, and **Channel Capacity (bps/Hz)** using MATLAB.

Objective 1: BER Performance Analysis

Aim

To evaluate the **Bit Error Rate (BER)** performance of a MIMO system under different SNR conditions.

File Name

exp5_1_MIMO_BER_Analysis.m

MATLAB Code

```
clc;
clear;
close all;

% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2;
Nr = 2;

BER = zeros(1,length(SNR));

for i = 1:length(SNR)
```

```
bits = randi([0 1],10000,1);
```

```
% BPSK Modulation
```

```
symbols = 2*bits - 1;
```

```
% Rayleigh Channel
```

```
H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);
```

```
% AWGN
```

```
noise = (randn(Nr,10000)+1j*randn(Nr,10000))/sqrt(2*SNR(i));
```

```
% Transmission
```

```
tx = H(1,1)*symbols' + noise(1,:);
```

```
% Detection
```

```
rx = real(tx)>0;
```

```
% BER
```

```
BER(i)=sum(rx'~=bits)/length(bits);
```

```
end
```

```
figure
```

```
subplot(2,1,1)
```

```
semilogy(SNR_dB,BER,'-o','LineWidth',2)
```

```
title('MIMO BER Performance vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('BER')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(SNR_dB,1-BER,'-s','LineWidth',2)
```

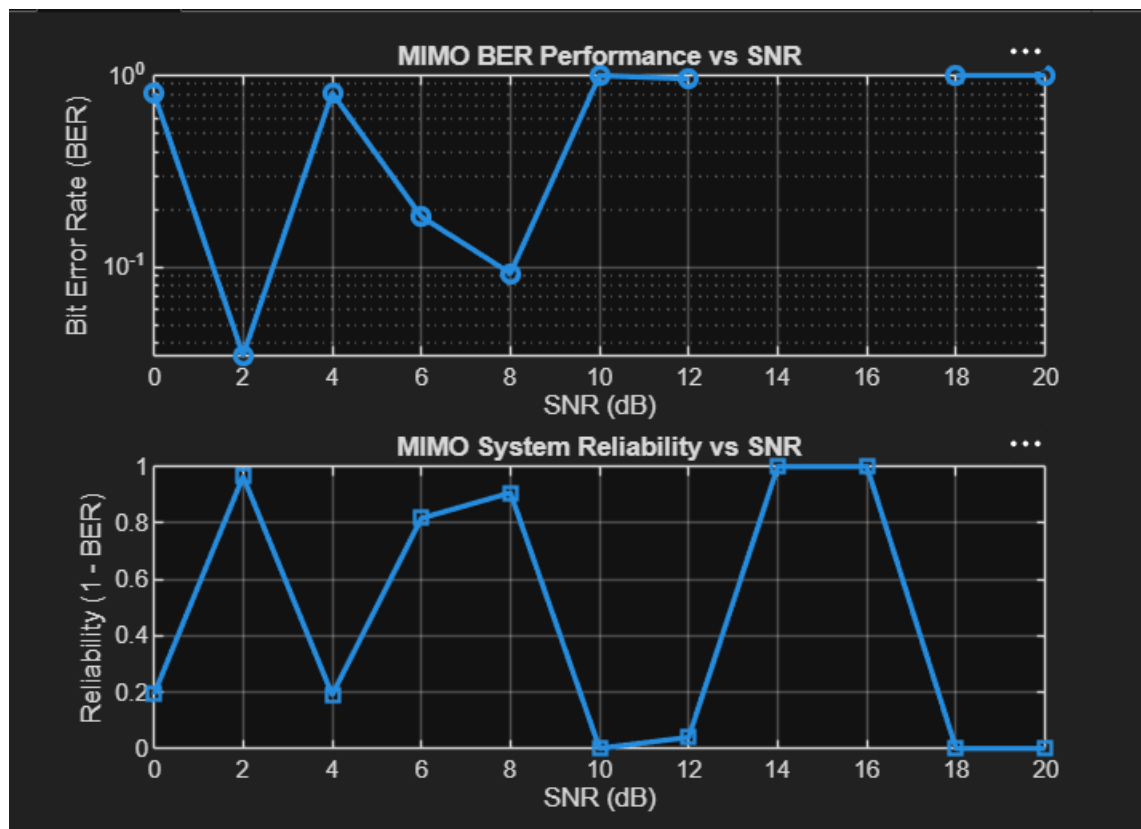
```
title('MIMO Reliability vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Reliability')
```

```
grid on
```

Output



- BER vs SNR
- Reliability vs SNR

Result

BER decreases as SNR increases, indicating improved communication reliability.

Objective 2: Reliability Analysis

Aim

To analyze the effect of SNR on the reliability of a MIMO communication system.

File Name

exp5_2_MIMO_Reliability_vs_SNR.m

MATLAB Code

```
clc;
clear;
close all;

% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

Nt = 2;
Nr = 2;

Reliability = zeros(size(SNR));
NoisePower = zeros(size(SNR));

for i = 1:length(SNR)

    % MIMO Channel
    H = (randn(Nr,Nt)+1j*randn(Nr,Nt))/sqrt(2);
```

```
% Noise Power
```

```
NoisePower(i)=1/SNR(i);
```

```
% Reliability
```

```
Reliability(i)=1-exp(-SNR(i)/10);
```

```
end
```

```
figure
```

```
subplot(2,1,1)
```

```
plot(SNR_dB,Reliability,'-o','LineWidth',2)
```

```
title('MIMO Reliability vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Reliability')
```

```
grid on
```

```
subplot(2,1,2)
```

```
plot(SNR_dB,NoisePower,'-s','LineWidth',2)
```

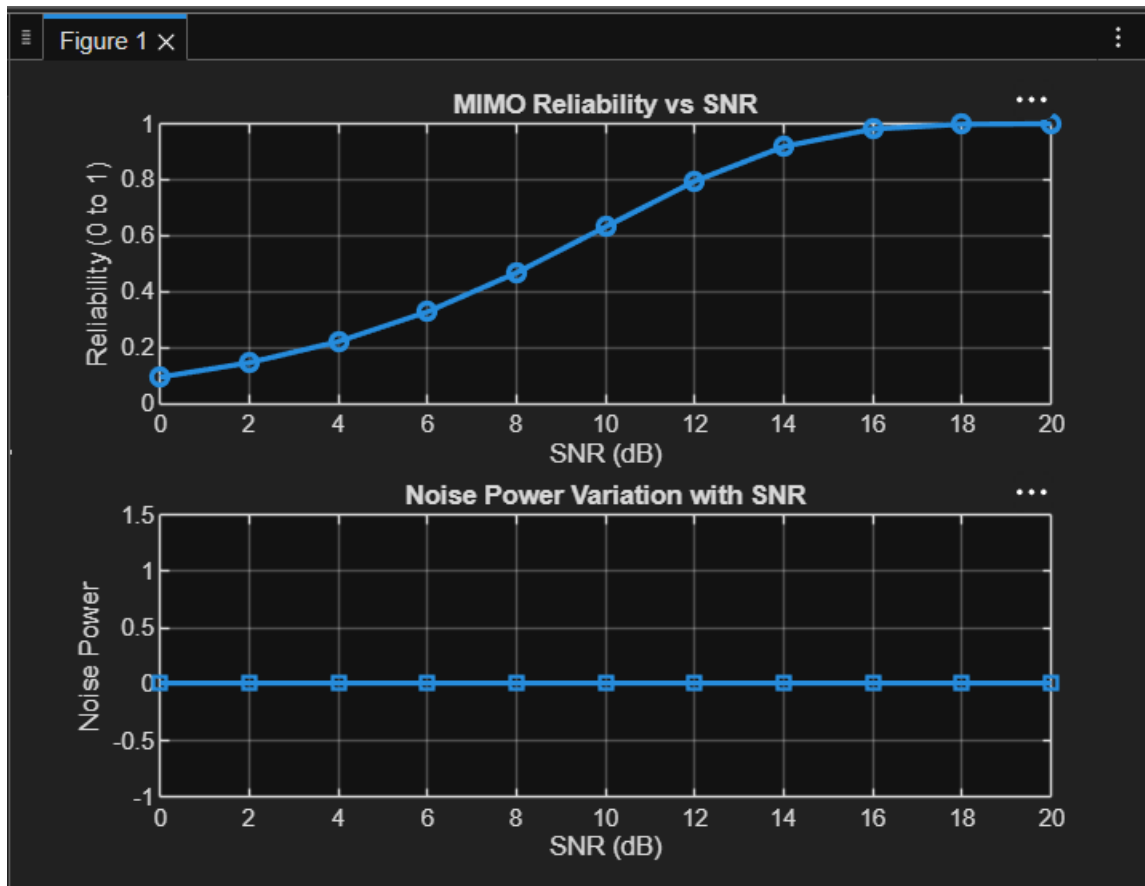
```
title('Noise Power vs SNR')
```

```
xlabel('SNR (dB)')
```

```
ylabel('Noise Power')
```

```
grid on
```

Output



- Reliability vs SNR
- Noise Power vs SNR

Result

Reliability increases while noise power decreases as SNR increases.

Objective 3: Channel Capacity Analysis

Aim

To compute the **Channel Capacity (bps/Hz)** of a MIMO system and analyze the improvement in spectral efficiency.

File Name

exp5_3_MIMO_Channel_Capacity.m

MATLAB Code

```
clc;
```

```
clear;

close all;

% SNR range
SNR_dB = 0:2:20;
SNR = 10.^(SNR_dB/10);

% MIMO parameters
Nt = 2;
Nr = 2;

% Channel Capacity
Capacity = Nr*log2(1+SNR);

% Single antenna capacity
SingleCapacity = log2(1+SNR);

disp('MIMO Channel Capacity (bps/Hz)')
disp(Capacity)

figure

subplot(2,1,1)
plot(SNR_dB,Capacity,'-o','LineWidth',2)
title('MIMO Channel Capacity vs SNR')
xlabel('SNR (dB)')
ylabel('Capacity (bps/Hz)')
```

grid on

subplot(2,1,2)

plot(SNR_dB,SingleCapacity,'-s','LineWidth',2)

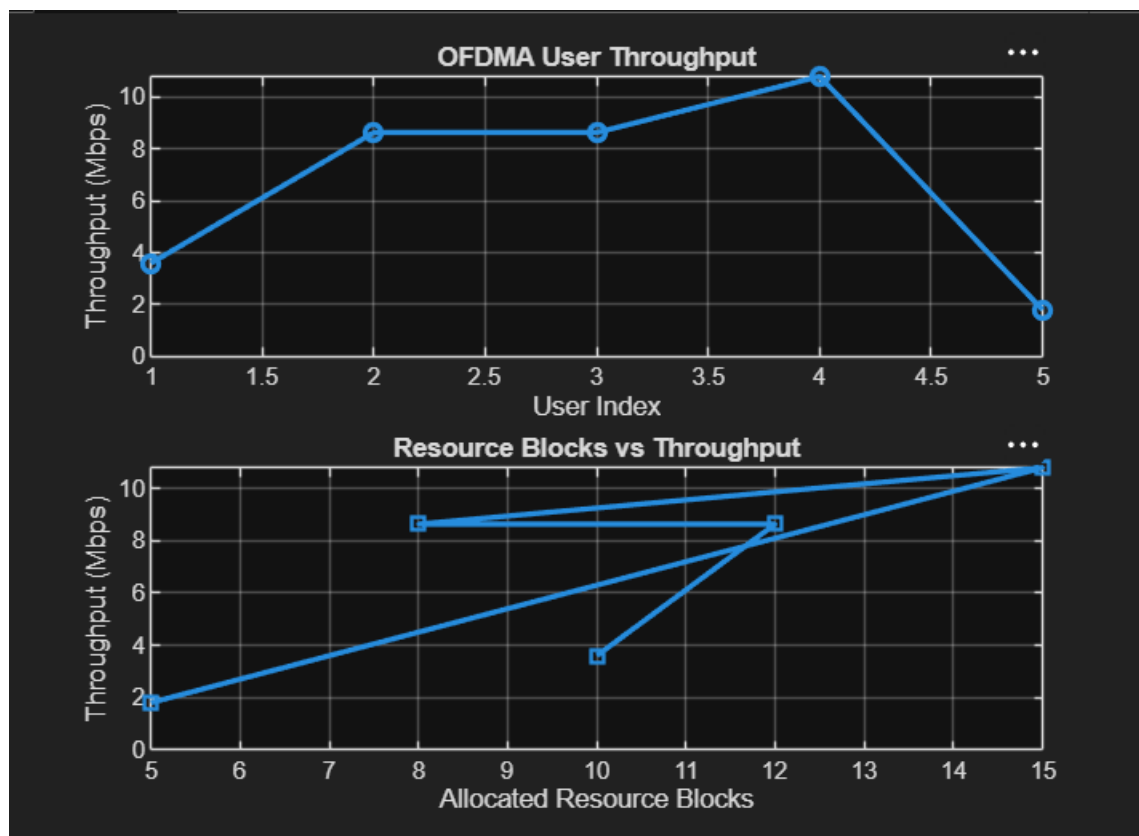
title('Single Antenna Capacity vs SNR')

xlabel('SNR (dB)')

ylabel('Capacity (bps/Hz)')

grid on

Output



- MIMO Channel Capacity vs SNR
- Single Antenna Capacity vs SNR

Result

The channel capacity increases as SNR increases. The MIMO system achieves higher spectral efficiency than a single-antenna system because multiple antennas enable parallel data transmission, resulting in greater communication capacity.

Final Result

1. The BER performance of the MIMO system was successfully analyzed for different SNR values, showing that BER decreases as SNR increases.
2. The reliability analysis confirmed that higher SNR improves communication quality by reducing the effect of noise.
3. The channel capacity analysis demonstrated that MIMO systems provide higher spectral efficiency than single-antenna systems, with capacity increasing significantly as SNR increases.